

AIM

Implementation of inter VLAN routing and dynamic routing in computer network using Cisco Packet Tracer.

OBJECTIVES

1. Understanding the use of VLAN in a computer network.
2. An overview on dynamic routing such as RIP between two different networks.
3. Configuring VLANs using layer 2 switch and verifying the connectivity between hosts using inter VLAN routing.
4. Verifying the connectivity between two different networks using dynamic routing protocol such as RIP.

THEORY

- VLAN (Virtual Local Area Network) is used to logically divide a single physical network into multiple broadcast domains.
- It reduces broadcast traffic, thereby improving overall network performance.
- VLANs enhance security by isolating different groups of users within the same network infrastructure.
- They simplify network management by allowing logical grouping of devices regardless of physical location.

Dynamic Routing (RIP) between two different networks.

- Dynamic routing allows routers to automatically exchange routing information without manual configuration.
- RIP (Routing Information Protocol) uses hop count as its routing metric to select the best path.
- Routers running RIP periodically share routing updates with neighbouring routers.
- RIP enables connectivity between different networks and adapts automatically to network changes.

OBSERVATIONS

VLAN-Inter-VLAN Routing

Fig. Inter-VLAN Routing configured between VLAN 10 and VLAN 20 using a Layer 2 switch and a router.
The setup successfully enables communication between hosts in different VLANs through inter-VLAN routing.

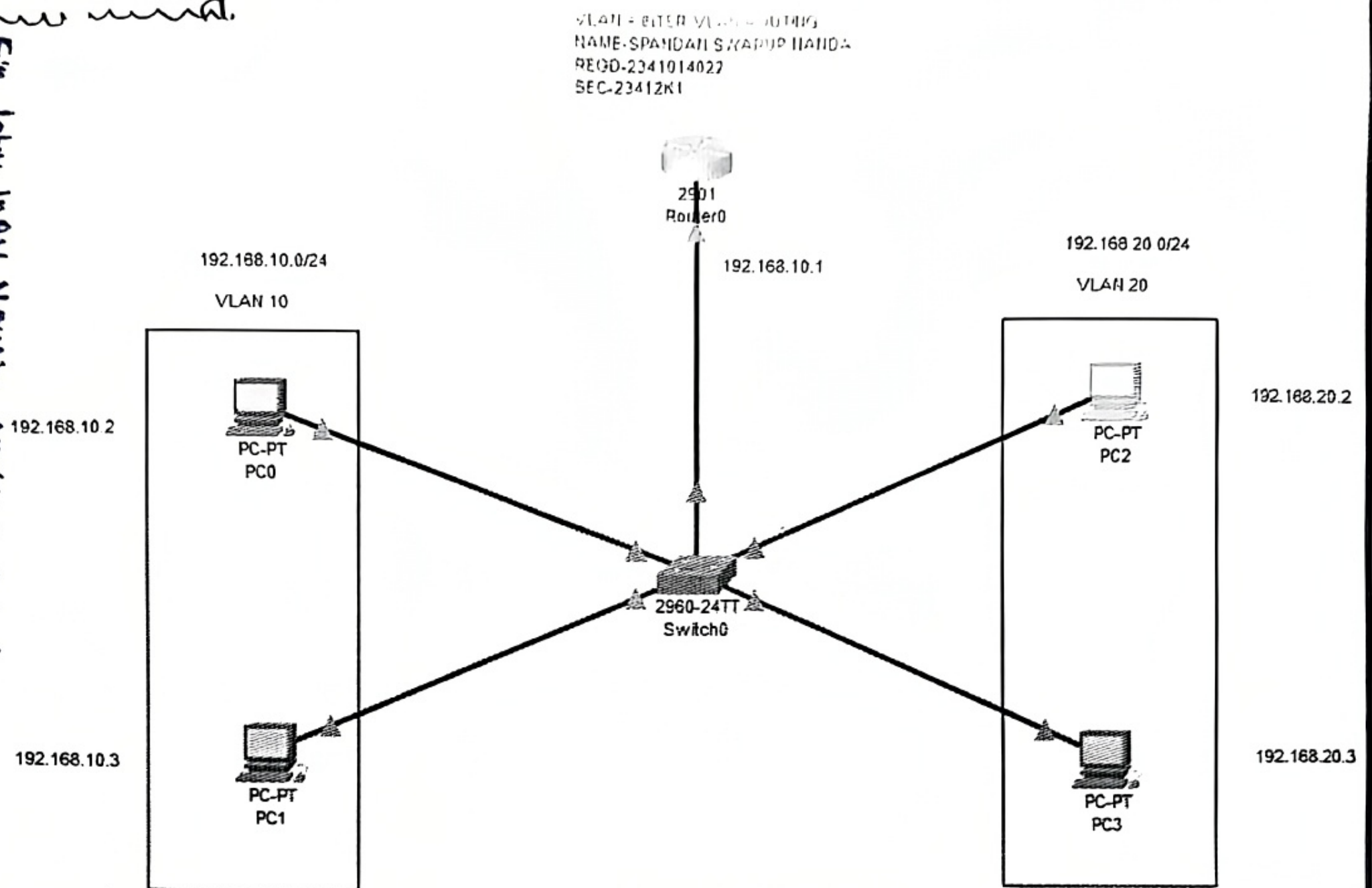


Fig. Router Subinterfaces configured for VLAN 10 and VLAN 20 using 802.1Q encapsulation.
The router-on-a-stick configuration enables inter-VLAN routing by assigning gateway IP addresses to each VLAN.

Router0

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface gig0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gig0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```



```

Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#show vlan brief
^
% Invalid input detected at '^' marker.

Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#show vlan brief
^
% Invalid input detected at '^' marker.

Switch(config)#show vlan brief
^
% Invalid input detected at '^' marker.

Switch(config)#
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   VLAN0010                active
20   VLAN0020                active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

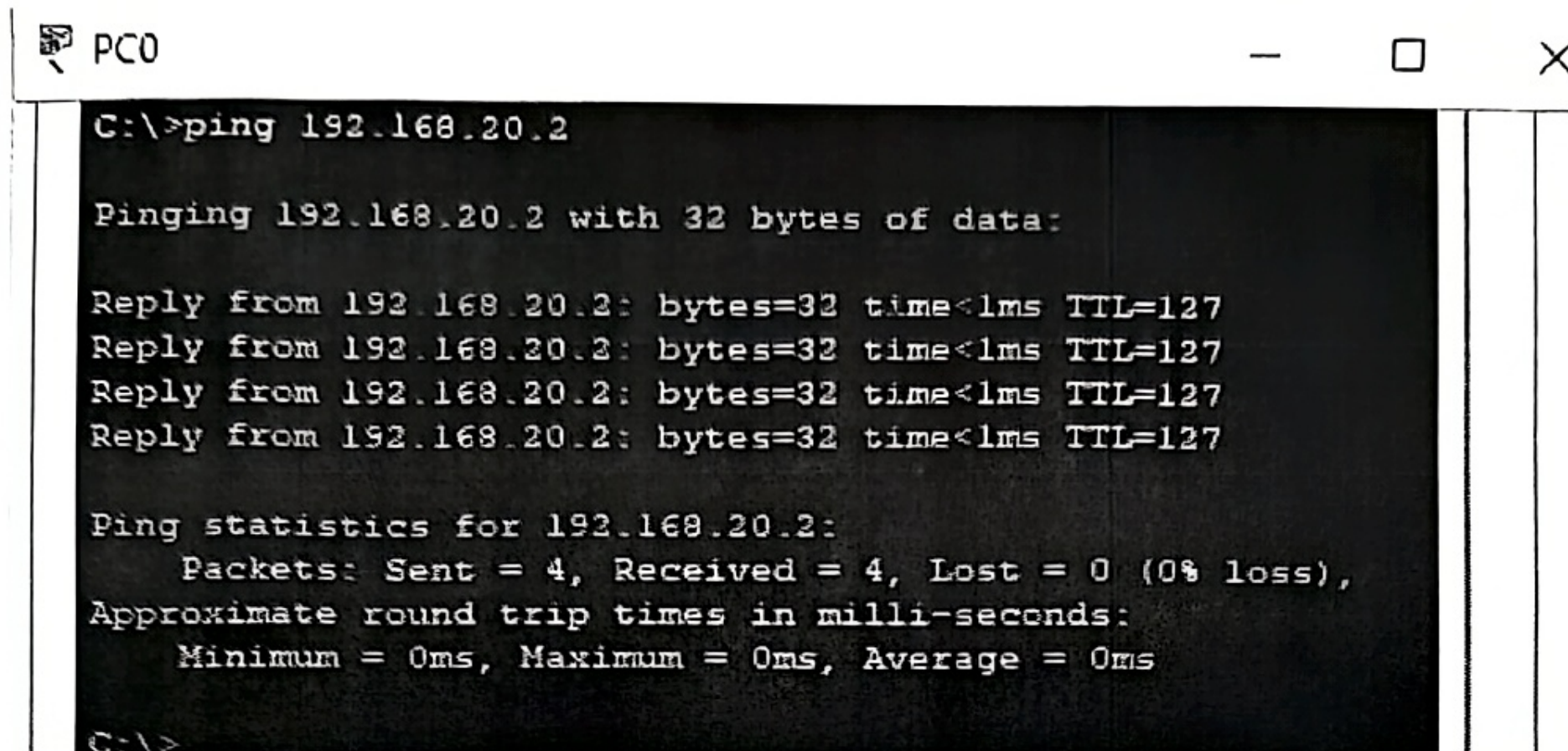
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/3 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

```

Fig. VLAN 10 and VLAN 20 created on a switch with access and trunk port configuration

↳ The switch is correctly configured to assign ports to respective VLANs and forward tagged traffic to the router via a trunk link.



```
PC0
C:\>ping 192.168.20.2

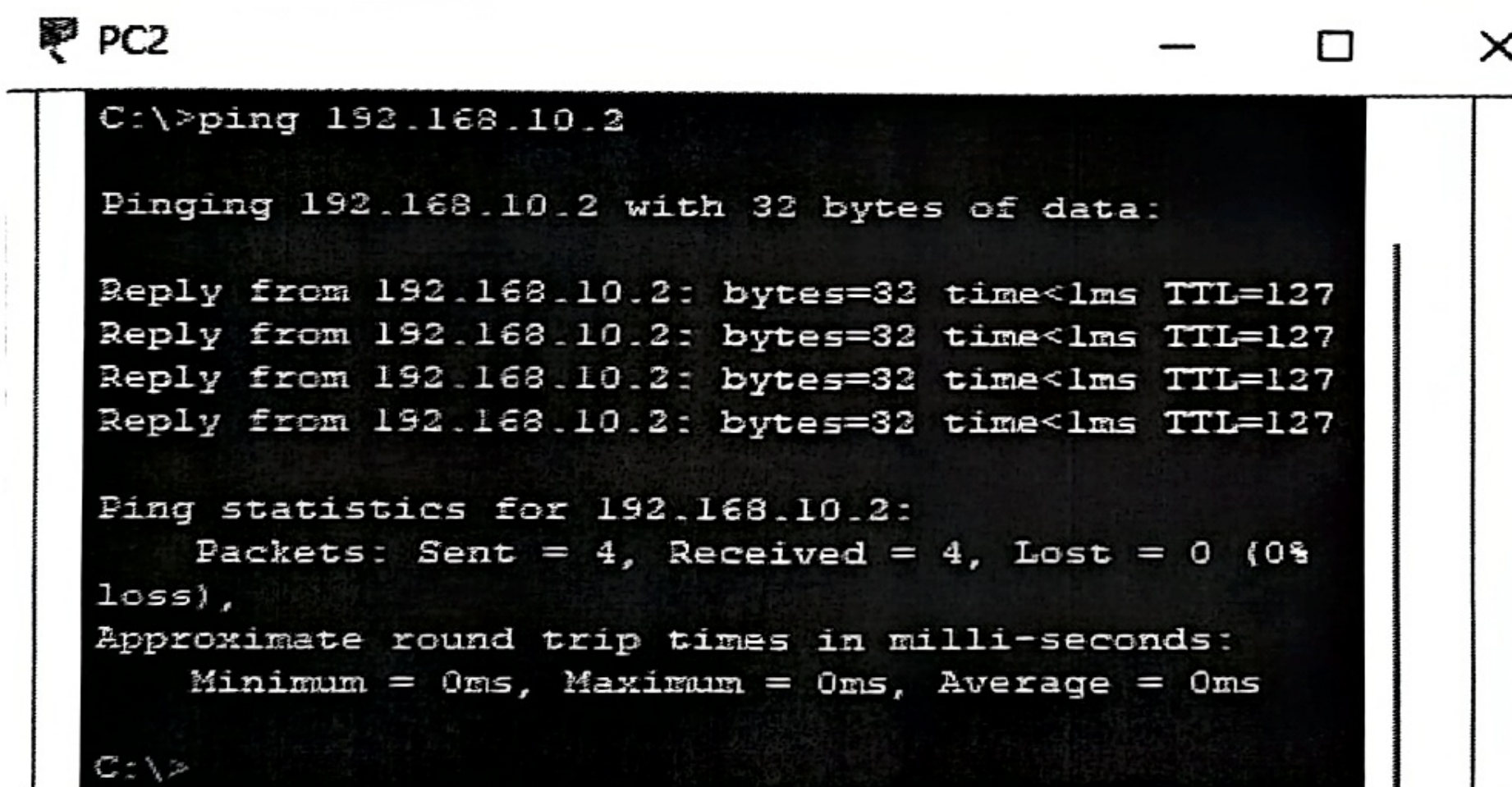
Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Fig. Successful ping verification between hosts in different VLANs using inter-VLAN routing.
↳ The successful ping confirms proper VLAN configuration and inter-VLAN routing through the router.



```
PC2
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
    loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Fig. Ping test from VLAN 20 host to VLAN 10 host verifying inter-VLAN connectivity.
↳ The successful ping response confirms bidirectional communication between VLANs via inter-VLAN routing.

(10) RIP (Routing Information Protocol)

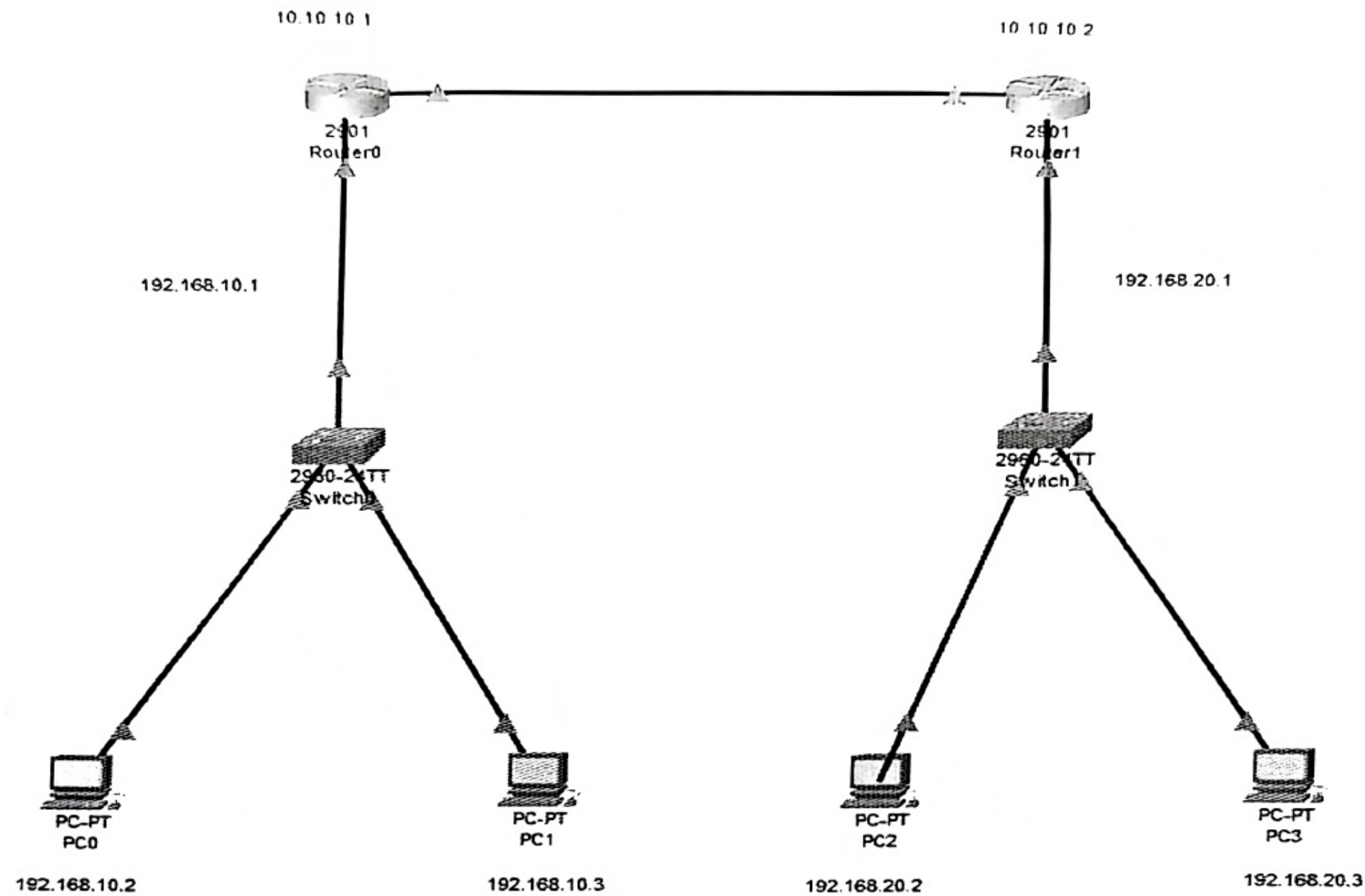


Fig. Dynamic routing configured using RIP to connect two different networks through multiple routers.

↳ RIP enables automatic route exchange, allowing successful communication between hosts in different networks.

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gig0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int gig0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.10.0
Router(config-router)#network 10.10.10.0
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/1
L       10.10.10.1/32 is directly connected, GigabitEthernet0/1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
R       192.168.20.0/24 [120/1] via 10.10.10.2, 00:00:06, GigabitEthernet0/1

Router#%IP-4-DUPADDR: Duplicate address 192.168.10.1 on GigabitEthernet0/0, sourced by 0002.176A.2E5A
```

Fig. RIP version 2 configured on Router0 for dynamic route exchange.
↳ The routing table confirms RIP: learned routes, enabling connectivity between different networks automatically.

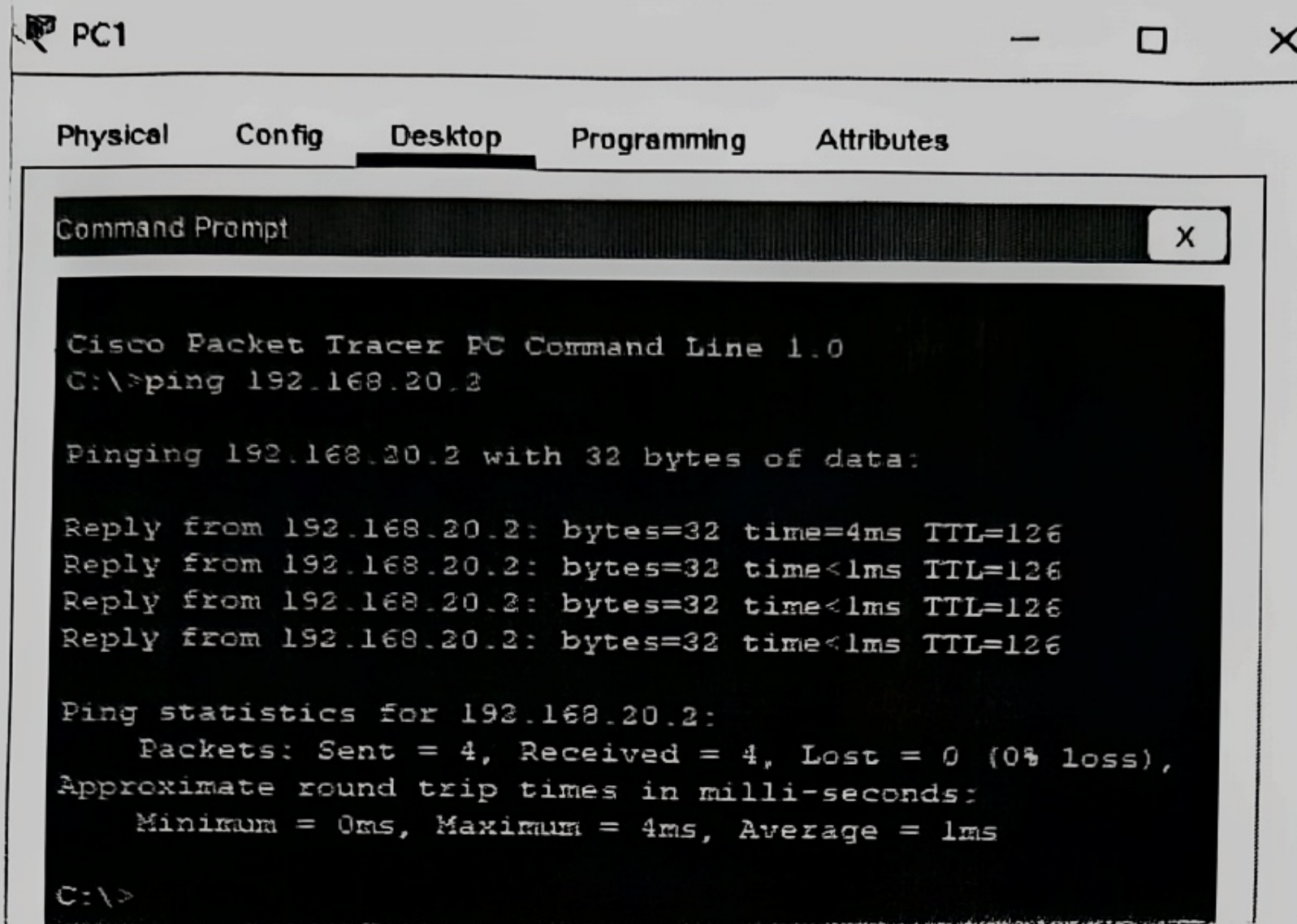


Fig. Ping test verifying connectivity between two different networks using RIP
↳ Successful ping confirms that dynamic routing via RIP is correctly configured and operational.

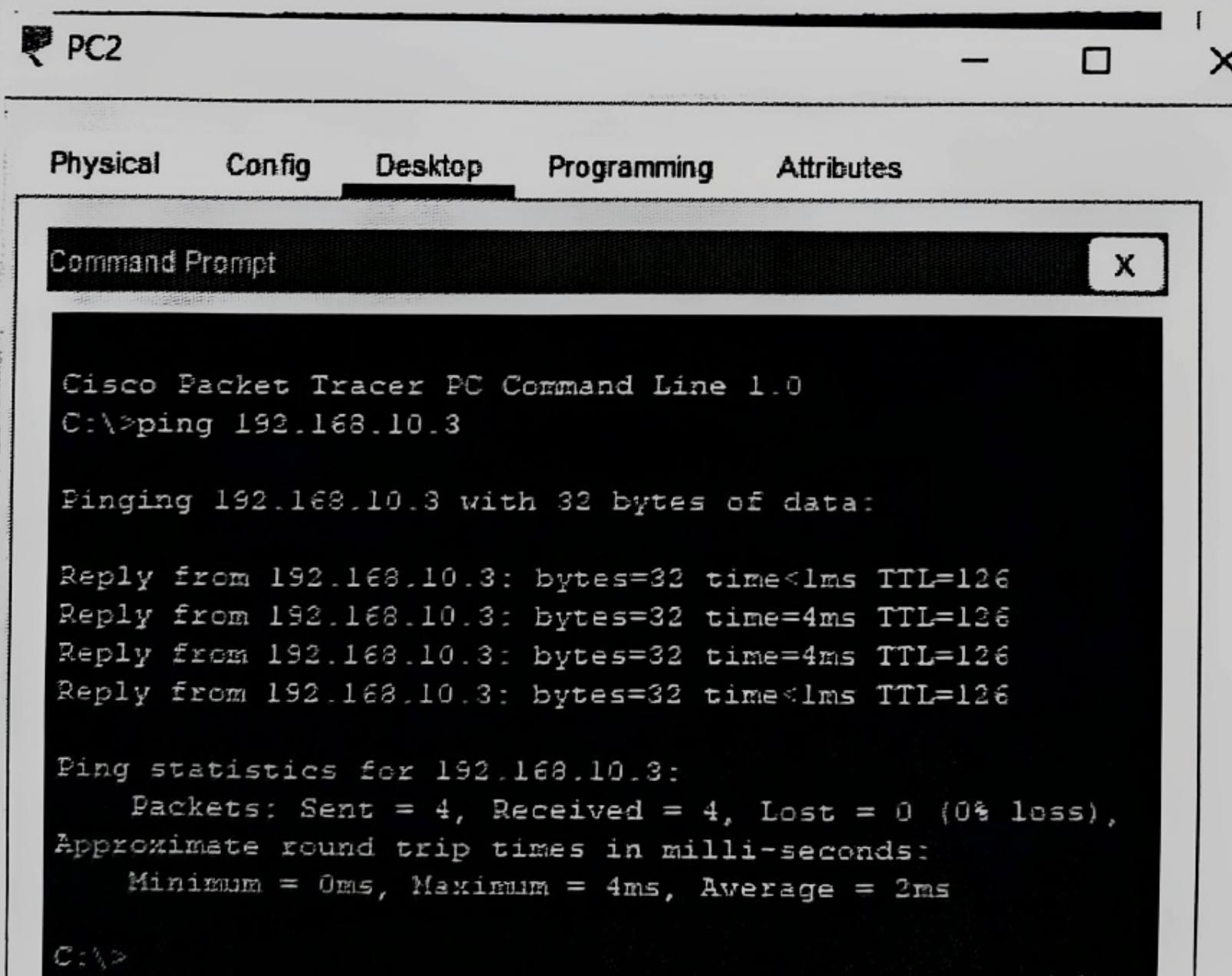


Fig. Ping Verification from a remote network host using RIP routing.
↳ The successful response shows that RIP has correctly learned routes and enabled inter-network communication.

ANALYSIS

(i) Inter VLAN Routing using Layer-2 Switch

- Multiple VLANs (VLAN 10 and VLAN 20) were created on a Layer 2 switch to logically segment the network.
- Switch ports were assigned to respective VLANs, enabling communication within the same VLAN only.
- A trunk link was configured between the switch and router to carry traffic of multiple VLANs.
- Router subinterfaces with 802.1Q encapsulation acted as gateways for each VLAN (Router-on-a-Stick).
- Successful ping between hosts of different VLANs verified correct inter-VLAN routing configuration.

(ii) Dynamic Routing using RIP

- Two different networks were interconnected using two routers configured with RIP version 2.
- IP addressing was correctly assigned to router interfaces and end devices as per the topology.
- RIP enabled automatic exchange of routing information between Router 0 and Router 1.
- The use of "no-auto-summary" ensured correct classless routing.
- Successful ping results between hosts of different networks confirmed proper RIP operation and end-to-end connectivity.

CONCLUSION

This experiment successfully demonstrated the use of VLANs for logical network segmentation and the implementation of inter-VLAN routing using a router. It also verified dynamic routing between different networks using RIP, confirming reliable communication, efficient routing, and proper connectivity across multiple networks in Cisco Packet Tracer.

EXERCISES

1. State the advantage of VLAN in computer networking.
2. How does a trunk help in inter VLAN routing?
3. State the similarity and differences between static and dynamic routing.
4. Differentiate OSPF routing from RIP routing?

(1) VLANs logically divide a physical network into multiple broadcast domains, reducing broadcast traffic and improving network performance, security, and manageability without requiring additional hardware.

(2) A trunk link carries traffic of multiple VLANs between the switch and router using VLAN tagging (802.1Q), allowing the router to route packets between different VLANs through subinterfaces.

(3) Differences

<u>Aspect</u>	<u>Static Routing</u>	<u>Dynamic Routing</u>
Configuration	Manually configured	Automatically learned
Adaptability	Does not adapt to changes	Automatically adapts.
Overhead	Low overhead	Higher overhead

Similarity:

Both methods are used to forward packets between different networks.

(4)

<u>Aspect</u>	<u>RIP</u>	<u>OSPF</u>
Metric	Hop Count	Cost (bandwidth-based)
Scalability	Limited (max 15 hops)	Highly scalable
Speed	Slower convergence	Faster convergence.

eg: RIP is suitable for small networks, while OSPF is preferred for large enterprise networks.